LAB 4 Individual Deliverable 2

Multi Segment Trajectory Optimization

Item	Value
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Repository	https://github.com/lizh-coder/Robotics-Integration-Team-Project-I

1. Minimum-snap trajectory over k segments

For minimum snap, r = 4. Euler-Lagrange gives a polynomial of degree 2r - 1 = 7 on each segment. Thus each segment has eight coefficients and k segments have 8k unknown coefficients.

Constraint accounting

The count below uses the endpoint coefficients of each segment. An intermediate waypoint occurs at the end of one segment and the beginning of the next; its known position is therefore written once for each adjoining segment.

Constraint type	Location and meaning	Scalar constraints
Waypoint position	There are k + 1 distinct waypoint positions. Written at every segment start and end, they give 2k endpoint equations.	2k
Free derivatives	At each of k - 1 intermediate waypoints, P', P'', P''' are free values selected by the QP. Each is applied to both incident segment endpoints.	6(k - 1)
Fixed endpoint derivatives	At the global start and end, P', P'', P''' are prescribed (usually zero).	6
Total	Matches the coefficient count.	2k + 6(k - 1) + 6 = 8k

Why the six free-derivative equations appear at an interior waypoint

The trajectory must be C^3, so P', P'', and P''' have the same value on both incident segments. These three values are not numerically prescribed: the optimizer selects them. Each free value is inserted once at the left segment endpoint and once at the right segment endpoint, giving six scalar endpoint equations per interior waypoint.

The resulting total is exactly 8k, so the piecewise seventh-degree polynomial is uniquely determined once the waypoint positions and endpoint derivatives are supplied.

2. General minimum-r-th-derivative case

For a cost integral of $(P^{(r)}(t))^2$ over k segments, Euler-Lagrange gives degree 2r - 1 on every segment. There are therefore 2r coefficients per segment and 2rk unknown coefficients overall.

Constraint type	Count
Waypoint positions at all segment endpoints	2k
Interior free derivatives of orders 1 through r - 1, each used at two segment endpoints	2(r - 1)(k - 1)
Fixed derivatives of orders 1 through r - 1 at the two global endpoints	2(r - 1)

Constraint type	Count
Total	$2k + 2(r - 1)(k - 1) + 2(r - 1) = 2rk$

At each interior waypoint there are r - 1 free derivative values, $P^{(1)}$ through $P^{(r - 1)}$. Each value is imposed at both incident segment endpoints, producing 2(r - 1) equations. At the optimum, the Euler-Lagrange stationarity conditions also imply matching derivatives $P^{(r)}$ through $P^{(2r - 2)}$. The count reduces to the minimum-snap result above when r = 4.

Reference

MIT VNAV Lab 4 Exercises, Individual Deliverable 2. https://vnav.mit.edu/labs_2023/lab4/exercises.html